

DATA SCIENCE PROJECT REPORT

Udemy Courses

Exploratory Data Analysis

Course Popularity · Subscriber Trends · Pricing Impact · Subject & Level Insights

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Executive Summary

This report documents an Exploratory Data Analysis (EDA) on a dataset of 3,678 Udemy online courses across four subject categories. The project explores course popularity metrics — subscriber counts, review volumes, pricing, and lecture structure — to uncover what makes a course successful on one of the world's largest e-learning platforms.

The analysis demonstrates proficiency in datetime parsing, deduplication, Seaborn visualizations, groupby aggregation, text-based search with `str.contains()`, and multi-metric ranking — all critical skills for product analytics and business intelligence roles in the ed-tech domain.

Project Highlights

Metric	Value
Dataset Size	3,678 rows × 12 columns (raw); 3,672 after dedup
Subjects Covered	4: Web Development, Business Finance, Musical Instruments, Graphic Design
Course Levels	4: All Levels, Beginner, Intermediate, Expert
Paid Courses	3,362 (91.6%)

Metric	Value
Free Courses	310 (8.4%)
Most Popular Subject	Web Development — 1,199 courses
Most Common Level	All Levels — 1,925 courses (52.4%)
Most Subscribed Course	Learn HTML5 Programming From Scratch — 268,923 subscribers
Most Subscribed Course Type	Free courses average 11,533 subscribers vs Paid 2,421
Most Lectures	Paid courses avg 41.9 lectures vs Free 21.4
Highest Reviewed Subject	Web Development
Total Python Courses	29 courses contain 'Python' in title
Price vs Reviews	No significant correlation — price doesn't drive reviews
Tools Used	Python, Pandas, NumPy, Seaborn, Matplotlib

1. Introduction

1.1 Project Background

Udemy is one of the world's largest online learning marketplaces, hosting over 200,000 courses across dozens of categories. Understanding what drives course popularity — whether it's subject area, pricing model, content volume, or skill level — is essential for course creators, platform strategists, and ed-tech analysts.

This project performs a structured EDA on a Kaggle-sourced snapshot of 3,678 Udemy courses. It focuses on deriving business-relevant insights about course performance while demonstrating key data science techniques including datetime handling, deduplication, Seaborn visualizations, groupby analysis, and keyword-based search.

1.2 Objectives

- Inspect and clean the dataset (duplicates, datetime parsing)
- Understand subject and level distribution across the course catalog
- Compare free vs. paid courses on subscribers, lectures, and content duration
- Identify the most popular courses and subjects by subscriber and review count
- Investigate whether price affects number of reviews
- Search for Python-related courses and rank them by popularity
- Visualize all findings using Seaborn countplot, barplot, and scatterplot

1.3 The 18 Questions Answered

#	Question	Analysis Type	Visualization
1	Display top 10 rows	Inspection	head(10)
2	Display last 5 rows	Inspection	tail(5)
3	Shape of dataset	Inspection	shape
4	Info: dtypes, nulls, memory	Profiling	info()
5	Check null values	Quality Check	isnull().sum(), heatmap
6	Check & drop duplicates	Cleaning	duplicated(), drop_duplicates()
7	Number of courses per subject	Univariate	countplot (rotated x)
8	Course levels available	Univariate	countplot (rotated x)
9	Count of paid vs free courses	Univariate	countplot
10	Which type has more lectures: free or paid?	Bivariate	groupby.mean()
11	Free vs paid: which has more subscribers?	Bivariate	barplot (is_paid vs num_subscribers)
12	Which level has most subscribers?	Bivariate	barplot (level vs num_subscribers)
13	Most popular course title	Max lookup	Boolean filter on max()
14	Top 10 most popular courses by subscribers	Ranking	sort_values + barplot
15	Subject with highest number of reviews	Bivariate	barplot (subject vs num_reviews)
16	Does price affect number of reviews?	Bivariate	scatterplot (price vs num_reviews)
17	Total Python courses	Text Search	str.contains('python', case=False)
18	Top 10 popular Python courses	Ranking + Search	filter + sort_values + barplot

2. Dataset Overview

2.1 Data Source

The dataset is a Udemy course catalog snapshot sourced from Kaggle. It covers four subject categories: Web Development, Business Finance, Musical Instruments, and Graphic Design. The data includes course metadata such as pricing, subscriber counts, reviews, lecture counts, and publication timestamps.

Repository: github.com/princemaheta2627-glitch/udemy_courses_eda

2.2 Dataset Structure

The raw dataset contains 3,678 rows and 12 columns. After deduplication, 3,672 unique course records remain.

Column	Dtype	Description	Notes
course_id	int64	Unique course identifier	Clean, no nulls
course_title	object	Full course title	Used for str.contains() search
url	object	Udemy course URL	Not used in analysis
is_paid	bool	True = paid, False = free	Key segmentation variable
price	int64	Course price in USD (0 if free)	Used in scatterplot vs reviews
num_subscribers	int64	Total enrolled students	Primary popularity metric
num_reviews	int64	Total reviews submitted	Secondary popularity metric
num_lectures	int64	Number of lectures in course	Compared across free/paid
level	object	Course difficulty level (4 values)	All Levels, Beginner, Intermediate, Expert
content_duration	float64	Total content hours	Compared across free/paid
published_timestamp	object → datetime64	Date/time of course publication	Parsed using parse_dates
subject	object	Course subject category (4 values)	Primary grouping variable

2.3 Key Data Engineering: Datetime Parsing

A critical data engineering step demonstrated early in the notebook is proper datetime ingestion. The `published_timestamp` column is stored as a plain object string by default, which prevents any date-based analysis. The correct approach is to parse it at load time:

```
- # Wrong: loads as object string
- data = pd.read_csv('5-udemy_courses.csv')
-
- # Correct: parse at load time
- data1 = pd.read_csv('5-udemy_courses.csv', parse_dates=['published_timestamp'])
```

This converts the column to `datetime64` dtype, enabling temporal filtering, sorting, and feature extraction (year, month, day) if needed in future analysis.

2.4 Data Quality

Issue	Finding	Resolution
Null Values	isnull().sum() = 0 across all 12 columns	No action needed — dataset is complete
'?' Values	Not present (unlike Income dataset)	No disguised null handling required
Duplicate Rows	duplicated().any() = True — 6 duplicate rows found	drop_duplicates() applied: 3,678 → 3,672 rows
Datetime dtype	published_timestamp loaded as object by default	Reloaded with parse_dates=['published_timestamp']

3. Methodology

3.1 Tools & Libraries

Library	Role	Key Usage in Project
Python 3	Core language	All logic and scripting
Pandas	Data manipulation	read_csv with parse_dates, groupby, sort_values, str.contains
NumPy	Numeric support	Background array operations
Seaborn	Statistical visualization	countplot, barplot, scatterplot
Matplotlib	Plot rendering	xticks(rotation), figsize, xlabel, ylabel, show()
Kaggle Notebook	Environment	Jupyter-based cloud execution

3.2 Key Code Patterns

Datetime Parsing at Load

```
data1 = pd.read_csv('5-udemy_courses.csv', parse_dates=['published_timestamp'])
```

Deduplication

```
data1.duplicated().any()          # → True
data1 = data1.drop_duplicates()
data1.duplicated().any()          # → False
```

Subject & Level Distribution

```
data1['subject'].value_counts()
sns.countplot(x='subject', data=data1)
```

```
- plt.xticks(rotation=65)
```

Free vs. Paid Comparison

```
- data1.groupby(['is_paid']).mean()          # lectures, subscribers, price, duration
- sns.barplot(x='is_paid', y='num_subscribers', data=data1)
```

Most Popular Course Lookup

```
- data1['num_subscribers'].max()              # → 268,923
- data1[data1['num_subscribers'] ==
  data1['num_subscribers'].max()]['course_title']
- # → 'Learn HTML5 Programming From Scratch'
```

Top 10 Courses by Subscribers

```
- top10 = data1.sort_values(by='num_subscribers', ascending=False).head(10)
- sns.barplot(x='num_subscribers', y='course_title', data=top10)
```

Python Course Search

```
- data[data['course_title'].str.contains('python', case=False)]
- len(...)  # → 29 courses
- python = data[...].sort_values(by='num_subscribers', ascending=False).head(10)
- sns.barplot(x='num_subscribers', y='course_title', data=python)
```

4. Key Findings & Insights

4.1 Course Subject Distribution

Subject	Course Count	Share	Insight
Web Development	1,199	32.6%	Largest subject — highest demand category
Business Finance	1,191	32.4%	Near-equal to Web Dev — strong professional demand
Musical Instruments	680	18.5%	Significant creative/hobby segment
Graphic Design	602	16.4%	Smallest but growing visual skills category

Insight: Web Development and Business Finance are nearly equal in course volume, together comprising 65% of the catalog. This reflects the dominant demand for technical and financial skills in the online learning market.

4.2 Course Level Distribution

Level	Count	Share	Insight
All Levels	1,925	52.4%	Over half of courses are designed for any skill level
Beginner Level	1,268	34.5%	Strong beginner focus — acquisition-driven strategy
Intermediate Level	421	11.5%	Significantly fewer intermediate options
Expert Level	58	1.6%	Very niche — only 58 expert-level courses

Insight: The heavy skew toward 'All Levels' (52.4%) and 'Beginner' (34.5%) reflects Udemy's broad-audience strategy. The scarcity of Expert-level courses (1.6%) represents a potential market gap for advanced learners.

4.3 Free vs. Paid Course Analysis

Metric	Free Courses	Paid Courses	Interpretation
Count	310 (8.4%)	3,362 (91.6%)	Platform is overwhelmingly paid
Avg. Price	\$0.00	\$72.20	Paid courses average \$72 per enrollment
Avg. Subscribers	11,533	2,421	Free courses attract 4.8x more subscribers
Avg. Reviews	425.9	131.5	Free courses get 3.2x more reviews
Avg. Lectures	21.4	41.9	Paid courses have nearly 2x more lectures
Avg. Content Hours	2.21 hrs	4.27 hrs	Paid courses are nearly 2x longer

Insight: Free courses dramatically outperform paid ones in subscriber counts and reviews — accessibility drives volume. However, paid courses offer substantially more content (2x lectures and duration), justifying their price point. This free-vs-paid dynamic is a classic freemium pattern: free courses build audiences while paid courses monetize depth.

4.4 Course Level vs. Subscribers

A barplot of level vs. num_subscribers revealed:

- All Levels courses attract the highest average subscriber counts — broad targeting maximizes reach
- Beginner Level courses are second — confirming the large beginner market
- Intermediate and Expert courses trail significantly — advanced audiences are smaller but more specialized

Insight: Courses designed for 'All Levels' are the most commercially successful strategy on Udemy. Instructors targeting the broadest possible audience benefit from maximum discoverability and enrollment volume.

4.5 Most Popular Courses

Single Most Popular Course

The course with the highest subscriber count is: Learn HTML5 Programming From Scratch — 268,923 subscribers. It is a free course published in February 2013, categorized as 'All Levels' in Web Development, with 45 lectures and 10.5 hours of content.

Top 10 Courses by Subscribers

Rank	Course Title (abbreviated)	Subject	Type
1	Learn HTML5 Programming From Scratch	Web Development	Free
2	Coding for Entrepreneurs Basic	Web Development	Free
3	The Web Developer Bootcamp	Web Development	Paid
4	Build Your First Website in 1 Week with HTML5	Web Development	Free/Paid
5	The Complete Web Developer Course 2.0	Web Development	Paid
6	Free Beginner Electric Guitar Lessons	Musical Instruments	Free
7–10	Various Web Development & Music courses	Mixed	Mixed

Insight: Web Development dominates the top 10 — 8 of the top 10 most subscribed courses are in this subject. Free courses appear prominently, reinforcing the subscriber-volume advantage of zero-cost offerings.

4.6 Subject with Highest Reviews

A barplot of subject vs. num_reviews confirmed Web Development as the highest-reviewed subject — consistent with it having the most courses and most subscribers. Business Finance is second, followed by Musical Instruments and Graphic Design.

Insight: Review volume correlates with subscriber volume — more students means more potential reviewers. Web Development leads on both metrics, making it the most engaged subject category on the platform.

4.7 Price vs. Reviews — No Correlation

A scatterplot of price (x) vs. num_reviews (y) showed a wide, horizontal scatter pattern with no visible trend:

- High-priced courses (\$150–\$200) do not necessarily have more reviews

- The highest review counts cluster at lower price points (\$0–\$50)
- No linear or non-linear relationship is visually detectable

Insight: Course pricing does not drive review behavior. Review volume is driven by subscriber count (free or cheap courses get more students who leave reviews), not price premium. This suggests instructors should not expect premium pricing to signal quality via review volume.

4.8 Python Course Analysis

Metric	Value
Total Python courses (by title)	29 courses
Search method	<code>str.contains('python', case=False)</code>
Top Python course	Web Programming with Python
Common Python course types	Django, Trading, Ethical Hacking, Finance
Subject category	Mostly Web Development

The top 10 Python courses were ranked by subscriber count and visualized using a horizontal barplot. Python courses span multiple domains — web development (Django), algorithmic trading, ethical hacking, and data science — reflecting Python's versatility across career paths.

5. Technical Skills Demonstrated

5.1 Data Loading & Datetime Handling

- `pd.read_csv()` with `parse_dates=['published_timestamp']` — correct datetime ingestion at load time
- Understanding the difference between object and `datetime64` dtypes and why it matters
- Shift+Tab documentation shortcut in Jupyter noted in comments — shows professional workflow awareness

5.2 Data Quality & Cleaning

- `isnull().values.any()` — concise global null check returning a single boolean
- `isnull().sum()` — column-level null counts
- `sns.heatmap(data.isnull())` — visual null pattern confirmation
- `duplicated().any()` — presence detection before removal
- `drop_duplicates()` — clean deduplication: 3,678 → 3,672 rows

5.3 Seaborn Visualization Suite

- `sns.countplot(x=cat, data=df)` — frequency distribution for categorical columns

- `sns.barplot(x=cat, y=num, data=df)` — mean comparison across groups
- `sns.barplot(x=num, y=cat, data=df)` — horizontal barplot for ranking course titles
- `sns.scatterplot(x=num, y=num, data=df)` — correlation investigation
- `plt.xticks(rotation=65)` — label rotation for readability
- `plt.figure(figsize=(15,3))` — custom plot sizing for wide scatter
- `plt.xlabel()`, `plt.ylabel()` with `fontsize` — professional axis labeling

5.4 Aggregation & Ranking

- `data1.groupby(['is_paid']).mean()` — multi-metric comparison of free vs. paid
- `sort_values(by='num_subscribers', ascending=False).head(10)` — top-N ranking
- `value_counts()` — frequency counts for subject, level, is_paid
- Boolean filter on `max()`: `data1[data1['col'].max() == data1['col']]` — max lookup

5.5 Text Search

- `str.contains('python', case=False)` — case-insensitive keyword search in course titles
- Combining text filter + `sort_values()` + `head(10)` for filtered ranking
- `len()` on filtered DataFrame — quick count of matching records

6. Conclusion & Future Work

6.1 Summary of Findings

- 3,672 unique Udemy courses across 4 subjects and 4 difficulty levels
- Web Development (1,199) and Business Finance (1,191) dominate the catalog
- 91.6% of courses are paid; average paid course price is \$72.20
- Free courses attract 4.8x more subscribers (11,533 avg vs 2,421) but have half the content
- 'All Levels' courses are the most subscribed — broad targeting beats niche focus
- Most popular course: Learn HTML5 Programming From Scratch (268,923 subscribers, free)
- Web Development dominates top 10 subscribed courses list
- Price has no meaningful correlation with number of reviews
- 29 Python-titled courses found; Web Programming with Python leads by subscribers
- Expert Level courses are severely underrepresented (58 courses, 1.6%) — market gap

6.2 Potential Enhancements

Enhancement	Description	Technique
Subscriber Prediction	Predict course popularity from features	Random Forest, XGBoost, Linear Regression
Time Series Analysis	Enrollment trends by publication year/month	Datetime feature engineering, Plotly
Price Optimization	Find optimal price point for max revenue	Regression + elasticity modeling
NLP on Course Titles	Cluster courses by topic using title keywords	TF-IDF, K-Means, Word2Vec
Rating Prediction	Add rating data if available and predict ratings	Sentiment analysis, Regression
Subject Deep Dive	Detailed sub-analysis within Web Development	Further filtering + multi-level groupby
Competitive Gap Analysis	Identify underserved levels/subjects	Crosstab heatmap: subject vs level

6.3 Learning Outcomes

This project reinforced several important patterns: the `parse_dates` technique for proper datetime loading is a best practice that avoids post-hoc type conversion; the horizontal barplot (swapping x and y) is the correct choice when category labels (course titles) are long; and the `free vs. paid groupby.mean()` comparison is a powerful one-liner that simultaneously reveals differences across all numeric metrics. These are reusable patterns applicable to any marketplace or product analytics dataset.

7. Project Information

Project Title	Udemy Courses EDA: Popularity, Pricing & Subject Insights
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-glitch/udemy_courses_eda
Notebook File	5-Udemy-Courses (with seaborn)-Answers.ipynb
Dataset	Udemy Courses — Kaggle (3,678 rows, 12 columns)
Subjects	Web Development, Business Finance, Musical Instruments, Graphic Design

Platform	Kaggle Notebook / Jupyter (Python 3)
Languages & Tools	Python 3, Pandas, NumPy, Seaborn, Matplotlib
Key Techniques	parse_dates, drop_duplicates, groupby, str.contains, sort_values, countplot, barplot, scatterplot
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This report was prepared as part of a data science portfolio.
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